

21.11.16 (E)

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam

Nov – Dec 2016

Max. Marks: 100

Duration: 3 hrs

Class: S.Y.B.Tech

Semester: III

Name of the Course: Computer Organization and architecture

Branch: Computer Engineering

Course Code: UCEC304

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Max. Marks
Q 1 (a)	Define Computer Organization and Computer Architecture with examples	05
Q 1 (b)	Explain Von Neumann model in brief with a neat diagram	05
Q 1 (c)	What is a bus? Explain different types of buses	10
OR		
Q 1 (c)	Describe in brief about PCI and SCSI bus	10
Q 2(a)	What are IEEE 754 formats? Represent 175.1875 in single and double precision formats	10
Q2 (b)	Solve $12(M) * 5(Q)$ using Booth's Recoding Technique	10
OR		
Q 2(b)	Draw flowchart for Restoring division and hence solve for divisor=13 and dividend=40	10
Q3 (a)	Explain the register organization of CPU in detail	10
OR		
Q3 (a)	Explain instruction cycle state diagram with a neat diagram	10
Q3 (b)	Write a short note on various types of addressing modes of processor	10
Q4 (a)	What is the concept of Cache Coherence? Also brief on interleaved and associative memories	10
OR		
Q4 (a)	Differentiate between SRAM and DRAM	10
Q4 (b)	Explain RAID levels 0-6 with diagrams for each level	10
Q5 (a)	What is DMA? How is programmed I/O and interrupt driven I/O different from it?	10
Q5 (b)	Explain Flynn's classification in detail	10
OR		
Q 5 (b)	What is meant by 6 stage instruction pipeline? Support your answer with a flowchart	10